

DEKELATRO

Game Design Document

Dekel Riess · Introduction to C# Final Project · 2025

Dekelatro is a console-based card game inspired by Balatro, built entirely in C# as a final project for an Introduction to C# course at Tiltan College of Arts. The game implements a full poker hand evaluation engine, a Joker-powered multiplier system, a tiered shop economy, Planet card upgrades, and a blind progression across seven Antes. This document covers every system as implemented in the codebase.

1. Overview

Field	Detail
Genre	Roguelike Deck-BUILDER (Poker)
Platform	Console — Windows (.NET 10)
Language	C# — no game engine
Context	Solo project — Final project for Introduction to C#
Win Condition	Defeat all 21 blinds across 7 Antes
Lose Condition	Run out of hands before meeting a blind's score target

2. Core Game Loop

Each run follows a repeating structure: face a blind, score points with hand plays, defeat the blind, visit the shop, then face the next blind.

Phase	Description
1. Draw	8 cards are drawn from a freshly shuffled 52-card deck.
2. Play or Discard	Choose 1–5 cards to play, or discard (up to 3 times per blind) to search for better combinations.
3. Score Preview	The score is calculated and displayed before you confirm. Press Y to lock in, F to cancel without spending a hand.
4. Repeat	Repeat until the blind target is met (win) or all 4 hand plays are exhausted (loss).
5. Shop	After defeating a blind, visit the shop to spend earned gold on Jokers and Planet cards.
6. Next Blind	Progress to the next blind. Score resets. Jokers and Planet levels persist.

3. Turn Structure & Player Controls

3.1 Per-Blind Resources

Resource	Amount	Notes
Hand size	8 cards	Drawn fresh from a full reshuffled deck at the start of each blind

Hands per blind	4	Each confirmed play costs one hand — exhaust all = game over
Discards per blind	3	Discard 1–5 cards; replacements are drawn immediately
Cards per play	1–5	Fewer than 5 cards triggers partial hand evaluation

3.2 Controls

Input	Action
Numbers (e.g. 0 2 4)	Select cards to play — score is previewed before confirming
Y (after preview)	Confirm play — score is added, hand count decreases
F (after preview)	Cancel — no hand is spent
D	Enter discard mode — enter card indexes to discard, or F to cancel
S	Sort hand by rank (descending)
A	Sort hand by suit
J	Open Joker menu — view, reorder, or sell Jokers
X	Open Consumable menu — use Planet cards from inventory slots
C	View remaining cards in deck, grouped by suit

4. Scoring System

4.1 Formula

Final Score = (Card Chips + Hand Bonus Chips + Joker Chips) x Chip Multiplier x (Hand Mult + Joker Mult)

- **Card Chips** — sum of rank values of cards that "count" toward the hand type.
- **Hand Bonus Chips** — flat chip bonus determined by hand type.
- **Joker Chips** — additive chips from active Jokers.
- **Chip Multiplier** — multiplicative modifier from Jokers (default 1.0x; Constellation grows this).
- **Hand Mult** — flat multiplier from hand type.
- **Joker Mult** — additive multiplier from active Jokers.

4.2 Which Cards Contribute Chip Value?

- **Pair, Three of a Kind, Four of a Kind, Full House** — only the matched cards contribute chips.
- **Straight, Flush, Straight Flush** — all 5 cards contribute chips.
- **High Card** — only the single highest card contributes chips.
- **Partial hands (<5 cards)** — matching cards score as their type; unmatched cards score as high card.

4.3 Hand Type Base Values

Hand Type	Bonus Chips	Bonus Mult	Eval Chips	Eval Mult
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Straight Flush	100	6	100	7
Four of a Kind	80	5	60	6
Full House	60	4	40	3
Flush	50	4	35	3
Straight	30	4	30	3
Three of a Kind	25	3	30	2
Two Pair	18	2	20	1
Pair	10	2	10	1
High Card	5	1	5	0

Both Bonus and Eval values are added together in the final calculation.

4.4 Score Preview

Before any play is committed, Dekelatro shows a full breakdown: card values, hand name, total chips, total mult, and final score. The player can then confirm or cancel without losing a hand. This transparency is an intentional design choice — players learn the math by seeing it every turn.

5. Blind Progression

The game spans 7 Antes, each with a Small, Big, and Boss Blind — 21 blinds total. Defeating all 21 wins the run. Score resets to zero after each blind; Jokers and hand levels persist.

Ante	Small Blind	Big Blind	Boss Blind	Rewards (S / B / Boss)
1	300	450	600	\$3 / \$5 / \$5
2	900	1,200	1,500	\$3 / \$5 / \$5
3	2,500	5,000	10,000	\$3 / \$5 / \$5
4	10,000	15,000	25,000	\$3 / \$5 / \$5
5	25,000	37,500	56,000	\$3 / \$5 / \$5
6	50,000	75,000	110,000	\$3 / \$5 / \$5
7	100,000	115,000	150,000	\$3 / \$5 / \$5

The steepest scaling jump is between Ante 2 and Ante 3 (1,500 to 2,500/5,000/10,000). Players who haven't built a working Joker synergy by Ante 3 will likely fail to progress.

6. Economy

6.1 Gold Sources

Source	Amount	Notes
Starting gold	\$4	Fixed — every run starts with \$4
Small Blind win	\$3	Awarded on defeat

Big Blind win	\$5	Awarded on defeat
Boss Blind win	\$5	Awarded on defeat
Golden Joker	+\$4	Passive — triggers on every blind defeat while owned
Trading Card Joker	+\$3	First single-card discard destroys the card and grants \$3
Sell Joker	\$1–\$4	Common: \$2 Uncommon: \$3 Rare: \$4

6.2 Gold Sinks

Sink	Cost	Notes
Common Joker	\$3–\$5	Most cost \$4
Uncommon Joker	\$5–\$8	
Rare Joker	\$7–\$10	
Planet Card	\$4	Flat price for all types
Shop Reroll	\$5+	Base \$5, increases by \$1 per reroll within the same shop visit

Gold is tight — especially early. Choosing between buying a Joker now versus saving for a reroll next visit is a consistent tension point.

7. Shop

The shop opens after every blind defeat and offers 3 items: Jokers and Planet cards. Items are randomly generated — the player can buy, reroll, view Jokers/consumables, or leave.

- **Tier odds:** Common 70%, Uncommon 25%, Rare 5%.
- **Duplicate protection:** The shop never shows a Joker the player already owns, and won't generate two identical items in the same roll.
- **Reroll cost:** \$5, +\$1 per reroll. Chaos the Clown makes the first reroll free per visit.
- **Joker cap:** Maximum 5 Jokers. Buying a 6th is blocked.
- **Consumable cap:** 2 slots. Planet cards are lost if slots are full.

8. Joker System

Jokers are persistent modifiers that survive across blinds. They can provide bonus chips, bonus mult, or a multiplicative chip multiplier — and some have passive economic effects. Players hold up to 5 Jokers simultaneously; they can be reordered and sold at any time from the Joker menu.

8.1 Common Jokers (Drop Rate: 70%)

Name	Price	Sell	Effect
Gros Michel	\$5	\$2	+15 Mult. 1-in-6 chance to break (removed permanently) after each blind win.
Misprint	\$4	\$2	Grants +1 to +20 Mult randomly on each hand played.
Jolly Joker	\$4	\$2	+8 Mult when the played hand contains at least a Pair.

Mad Joker	\$4	\$2	+10 Mult when the played hand is exactly Two Pair (4 cards played).
Crazy Joker	\$4	\$2	+12 Mult when the played hand is a Straight (5 cards played).
Zany Joker	\$4	\$2	+12 Mult when the played hand is Three of a Kind (3 cards played).
Odd Todd	\$3	\$2	+31 Chips for each odd-ranked card played (A, 3, 5, 7, 9).
Even Steven	\$3	\$2	+4 Mult for each even-ranked card played (2, 4, 6, 8, 10).
Mystic Summit	\$4	\$2	+15 Mult if the player has 0 discards remaining in the current blind.
Chaos the Clown	\$4	\$2	The first shop reroll each visit costs \$0 instead of \$5.
Golden Joker	\$4	\$2	Earn +\$4 every time a blind is defeated.

8.2 Uncommon Jokers (Drop Rate: 25%)

Name	Price	Sell	Effect
The Mask	\$7	\$3	+20 Mult when the played hand contains no face cards (Jack, Queen, King).
Fibonacci	\$7	\$3	+8 Mult for each Fibonacci-ranked card played (A, 2, 3, 5, 8).
Trading Card	\$5	\$2	If the first discard of a blind is exactly 1 card, that card is destroyed and you gain \$3.
Constellation	\$8	\$3	Each time a Planet card is used from inventory, this Joker permanently gains +0.1x chip multiplier.

8.3 Rare Jokers (Drop Rate: 5%)

Name	Price	Sell	Effect
Pi Man	\$10	\$4	If the played hand contains an Ace, a 3, and a 4, the total chip count is raised to the power of 3.
Void Portal	\$7	\$3	Every 3rd discard in the run transforms the discarded cards into random cards of the same suit.

8.4 Joker Design Notes

- Chips and mult from Jokers add together across all owned Jokers; chip multipliers multiply together — Constellation becomes exponentially more powerful with each Planet used.
- Gros Michel's break mechanic creates tension: a powerful joker may vanish at the worst moment, forcing the player to hedge.
- Mystic Summit rewards full restraint — saving all 3 discards through an entire blind for a flat +15 Mult.
- Pi Man is niche by design: it does nothing unless the player deliberately holds Ace, 3, and 4, making it a build-defining pickup.
- Void Portal turns discards into a resource: the more aggressively you discard, the faster it fires.

9. Planet Cards

Planet cards are consumable items that permanently level up a specific hand type for the rest of the run. They cost \$4 in the shop and are stored in 2 consumable slots. Using a Planet card also triggers Constellation's multiplier growth. Hand level values persist across blinds but reset between runs.

Planet Card	Levels Up
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Pair Planet	Pair
Two Pair Planet	Two Pair
Three of a Kind Planet	Three of a Kind
Straight Planet	Straight
Flush Planet	Flush
Full House Planet	Full House
Four of a Kind Planet	Four of a Kind
Straight Flush Planet	Straight Flush

10. Card System

10.1 Deck Composition

Standard 52-card deck: 4 suits (Spades, Hearts, Diamonds, Clubs) x 13 ranks (2 through Ace). The deck is shuffled at the start of each blind using a Fisher-Yates shuffle. After a blind is won the deck is rebuilt fresh and reshuffled for the next blind.

10.2 Card Chip Values

Cards	Chip Value
2 through 10	Face value (2–10)
Jack	11
Queen	12
King	13
Ace	14

10.3 Suit Colour Coding

Console colours differentiate suits: Hearts = Red, Diamonds = Yellow, Spades = Grey, Clubs = Blue.

10.4 Card Enchantments (Groundwork)

The Card class includes an Enchantment list, laying the technical foundation for future modifiers (e.g. Foil, Holographic, Polychrome effects from the source material). In the current version, enchantments are not exposed through gameplay.

11. Win & Loss Conditions

Outcome	Trigger	Result
Win	All 21 blinds defeated	"You beat all Blinds! You win!" — game exits.
Lose	0 hands left, score below target	"Game Over!" — game exits.

12. Design Reflection

Dekelatro was a learning project, but several choices were deliberate design decisions rather than incidental implementations:

- **Score preview before committing** — Showing chips x mult x final score before a play is locked in makes the math transparent. Players learn hand values by reading them every turn, not by memorising a chart.
- **Partial hand support** — Playing fewer than 5 cards is fully scored, which opens niche builds. Odd Todd, for example, rewards playing just odd-ranked cards rather than always filling a full hand.
- **Joker variety as the build driver** — 18 Jokers across 3 tiers means most runs feel different. The tight economy forces players to commit to a synergy rather than picking up everything available.
- **Gros Michel as risk design** — The strongest common Joker has a 1-in-6 chance to break after each blind win. Players must weigh sustained power against the chance of losing it at a critical moment.
- **Constellation + Planet synergy loop** — Planet cards level up a hand type AND grow Constellation's multiplier. This creates a deliberate positive feedback loop that rewards stacking both.
- **Escalating reroll cost** — Prevents trivially fishing for the perfect Joker. Every reroll is a meaningful decision because the next one will be more expensive.